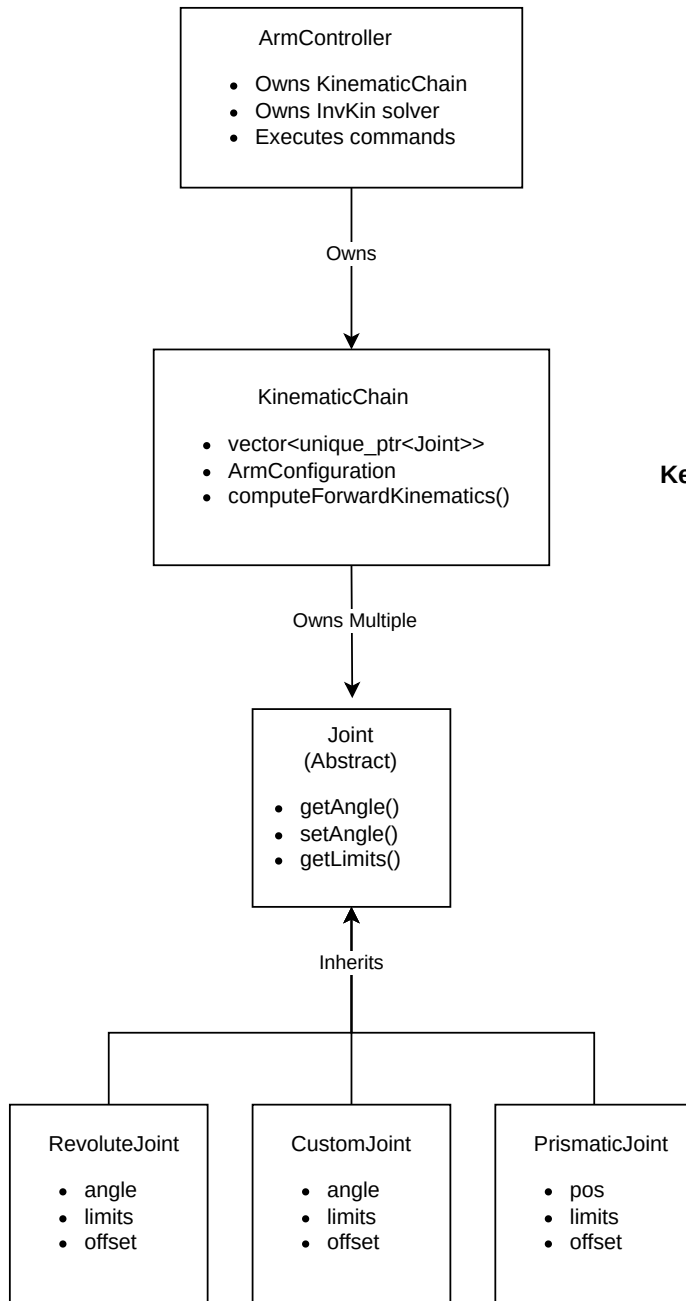
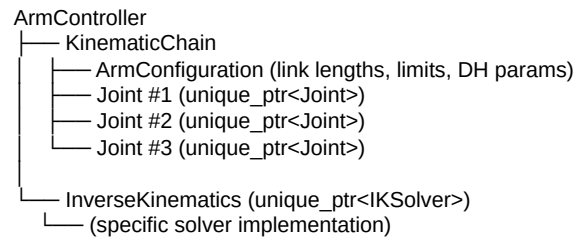




Composition View: Ownership



Key Responsibilities (SRP - Single Responsibility Principle)

Joint

- > Knows: its own state (angle, limits)
- > Does: validates ranges, stores position

KinematicChain

- > Knows: arm geometry, all joint states
- > Does: forward kinematics, manages joint collection

InverseKinematics

- > Knows: how to map XYZ -> joint angles
- > Does: solves IK, checks reachability

ArmCommand

- > Knows: what motion is requested
- > Does: encapsulates command data, validates itself

ArmController

- > Knows: nothing about math, delegates
- > Does: orchestrates components, executes commands

